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Contribution Journal

Brian – Standard Scaling

I ran the standard scaling code cell, which handled numerical features that can be scaled using the mean and standard deviation. In this lab, I was also able to review the scaling principle on Age and Fare, why features with different ranges can affect some machine learning methods, and how to inspect the transformed output. I also added comments to my code cell, which helped me understand what each part of my code did. Also, the team ran the code cells together to learn all the parts together.

Toan – Median Imputation for Missing Values

I took the lead on the code cell to fill missing values with the median. The team reviewed why missing data can cause problems during analysis or model training and how the median provides a typical middle value that is less affected by extreme values than the mean. I also reviewed the missing-value handling process, checked the result after the cell ran, and helped add code comments. Furthermore, the team ran the code cells together to learn and understand all the parts collectively.

Montserrat – Categorical Encoding

I took the lead on the categorical-encoding code cell. The team reviewed why text categories cannot always be used directly by machine-learning algorithms and how encoding represents

categories numerically. The cell used the dataset's categorical information to create an encoded data frame (encoded). The team checked the resulting columns and discussed how encoding makes categorical information usable in later steps of a machine-learning workflow. I also participated in code comments and shared a review of the results.

Shared Teamwork

Although each member had a main code cell, the work was collaborative. All three team members ran each code cell together, reviewed the outputs, discussed what the code was doing, and worked on comments explaining the steps. We did not treat a cell as complete just because it executed; we also discussed what its output meant and how it connected to the next part of data preparation. This shared review helped everyone understand all three tasks instead of learning only the cell assigned to them.